

Convexity, Constraints, and Optimality Conditions

Week 2 Lecture

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Learning goals

By the end of the lecture, you should be able to:

- recognize convex sets and convex functions;
- use first- and second-order characterizations of convexity;
- relate strong convexity, smoothness, and the Polyak–Lojasiewicz condition;
- formulate constrained problems and identify active constraints;
- state and apply the KKT conditions;
- derive weak duality and interpret dual multipliers as sensitivities.

Why convexity matters

For a general differentiable objective, $\nabla f(x) = 0$ identifies minima, maxima, and saddle points. Convexity removes this ambiguity.

Central consequence

For a differentiable convex function, every stationary point is a global minimizer.

Convexity also gives computable global bounds, meaningful dual problems, and convergence guarantees for first-order methods.

Convex combinations

Definition 1 (Convex combination)

Given $x_1, \dots, x_m \in \mathbb{R}^d$, a convex combination is

$$x = \sum_{i=1}^m \lambda_i x_i, \quad \lambda_i \geq 0, \quad \sum_{i=1}^m \lambda_i = 1.$$

The coefficients describe a weighted average. For two points,

$$(1 - t)x + ty, \quad t \in [0, 1],$$

traces the line segment from x to y .

Convex sets

Definition 2 (Convex set)

A set $C \subseteq \mathbb{R}^d$ is convex if

$$(1 - t)x + ty \in C \quad \text{for every } x, y \in C \text{ and } t \in [0, 1].$$

Examples include affine subspaces, half-spaces, norm balls, boxes, and the probability simplex

$$\Delta_d = \{x \in \mathbb{R}^d : x_i \geq 0, \mathbf{1}^\top x = 1\}.$$

The union of convex sets need not be convex. Their intersection is always convex.

Intersection preserves convexity: proof step 1

Theorem 3

For any family of convex sets $\{C_\alpha\}_{\alpha \in A}$, the intersection $C = \bigcap_{\alpha \in A} C_\alpha$ is convex.

Step 1. Choose arbitrary $x, y \in C$ and $t \in [0, 1]$.

Intersection preserves convexity: proof step 2

Theorem 4

For any family of convex sets $\{C_\alpha\}_{\alpha \in A}$, the intersection $C = \bigcap_{\alpha \in A} C_\alpha$ is convex.

Step 1. Choose arbitrary $x, y \in C$ and $t \in [0, 1]$.

Step 2. Since $x, y \in C$, we have $x, y \in C_\alpha$ for every $\alpha \in A$.

Intersection preserves convexity: proof step 3

Theorem 5

For any family of convex sets $\{C_\alpha\}_{\alpha \in A}$, the intersection $C = \bigcap_{\alpha \in A} C_\alpha$ is convex.

Step 1. Choose arbitrary $x, y \in C$ and $t \in [0, 1]$.

Step 2. Since $x, y \in C$, we have $x, y \in C_\alpha$ for every $\alpha \in A$.

Step 3. Convexity of each C_α gives $(1 - t)x + ty \in C_\alpha$ for every α . Hence $(1 - t)x + ty \in C$.

□

Convex, strictly convex, and strongly convex functions

Let C be convex and $f : C \rightarrow \mathbb{R}$.

Definition 6 (Convexity)

$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$ for all $x, y \in C$ and $t \in [0, 1]$.

Definition 7 (Strict convexity)

The inequality is strict for $x \neq y$ and $t \in (0, 1)$.

Definition 8 (μ -strong convexity)

For $\mu > 0$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y) - \frac{\mu}{2}t(1-t) \|x - y\|^2.$$

Strong convexity implies strict convexity, which implies convexity. The converses fail.

Examples separate the three notions

Function on $\mathbb{R}$	Convex	Strictly convex	Strongly convex
$ x $	yes	no	no
x^4	yes	yes	no on $\mathbb{R}$
$\frac{1}{2}x^2$	yes	yes	1-strongly convex
e^x	yes	yes	no on $\mathbb{R}$
constant function	yes	no	no

For x^4 , the second derivative $12x^2$ approaches zero near the origin, so no uniform positive curvature parameter exists.

First-order characterization of convexity

Theorem 9

Let C be open and convex, and let $f : C \rightarrow \mathbb{R}$ be differentiable. Then f is convex if and only if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \text{for all } x, y \in C.$$

Thus every tangent hyperplane globally underestimates a differentiable convex function.

First-order characterization: proof step 1

Assume f is convex. Fix $x, y \in C$ and define $z_t = x + t(y - x)$.

Step 1. Convexity gives

$$f(z_t) \leq (1 - t)f(x) + tf(y), \quad t \in (0, 1].$$

First-order characterization: proof step 2

Assume f is convex. Fix $x, y \in C$ and define $z_t = x + t(y - x)$.

Step 1. $f(z_t) \leq (1 - t)f(x) + tf(y)$.

Step 2. Rearranging and dividing by $t > 0$ yields

$$\frac{f(x + t(y - x)) - f(x)}{t} \leq f(y) - f(x).$$

First-order characterization: proof step 3

Assume f is convex. Fix $x, y \in C$ and define $z_t = x + t(y - x)$.

Step 1. $f(z_t) \leq (1 - t)f(x) + tf(y)$.

Step 2. $[f(x + t(y - x)) - f(x)]/t \leq f(y) - f(x)$.

Step 3. Letting $t \downarrow 0$ gives

$$\langle \nabla f(x), y - x \rangle \leq f(y) - f(x),$$

which proves the necessary direction.

First-order characterization: proof step 4

Assume the gradient inequality holds. Let $z = (1 - t)x + ty$.

Step 1. Apply it at z with comparison points x and y :

$$f(x) \geq f(z) + \langle \nabla f(z), x - z \rangle, \quad f(y) \geq f(z) + \langle \nabla f(z), y - z \rangle.$$

Step 2. Multiply by $1 - t$ and t , then add. The inner-product terms cancel because $(1 - t)(x - z) + t(y - z) = 0$.

Step 3. Therefore $(1 - t)f(x) + tf(y) \geq f(z)$. Hence f is convex. □

Second-order characterization

Theorem 10

Let C be open and convex, and let $f \in C^2(C)$. Then

$$f \text{ is convex} \iff \nabla^2 f(x) \succeq 0 \quad \forall x \in C.$$

Moreover, $\nabla^2 f(x) \succeq \mu I$ for all x implies μ -strong convexity.

The Hessian criterion converts a global inequality into a local matrix test.

Second-order characterization: proof step 1

Assume f is convex. Fix $x \in C$ and a direction d . Define $\phi(t) = f(x + td)$.

Step 1. Restricting a convex function to a line preserves convexity, so ϕ is convex near $t = 0$.

Second-order characterization: proof step 2

Assume f is convex and $\phi(t) = f(x + td)$.

Step 1. ϕ is convex.

Step 2. A twice differentiable one-dimensional convex function satisfies $\phi''(0) \geq 0$. By the chain rule,

$$\phi''(0) = d^\top \nabla^2 f(x) d \geq 0.$$

Since this holds for every d , $\nabla^2 f(x) \succeq 0$.

Second-order characterization: proof step 3

Assume now that $\nabla^2 f(z) \succeq 0$ everywhere. Fix x, y and set $d = y - x$.

Step 1. Taylor's theorem with integral remainder gives

$$f(y) = f(x) + \nabla f(x)^\top d + \int_0^1 (1-t) d^\top \nabla^2 f(x+td) d \, dt.$$

Second-order characterization: proof step 4

Assume $\nabla^2 f(z) \succeq 0$ everywhere and $d = y - x$.

Step 1. Use the integral Taylor formula.

Step 2. The integrand is nonnegative, so

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x).$$

Step 3. The first-order characterization proves convexity. Replacing $\nabla^2 f \succeq 0$ by $\nabla^2 f \succeq \mu I$ adds $\frac{\mu}{2} \|y - x\|^2$ and proves strong convexity. $\square$

Jensen's inequality

Theorem 11 (Jensen)

If f is convex, $x_i \in \text{dom } f$, $\lambda_i \geq 0$, and $\sum_i \lambda_i = 1$, then

$$f\left(\sum_{i=1}^m \lambda_i x_i\right) \leq \sum_{i=1}^m \lambda_i f(x_i).$$

Probabilistic form: $f(\mathbb{E}X) \leq \mathbb{E}f(X)$ whenever both sides exist.

Jensen's inequality: proof step 1

Step 1. The case $m = 2$ is exactly the definition of convexity.

Jensen's inequality: proof step 2

Step 1. The result holds for $m = 2$.

Step 2. Assume it holds for $m - 1$ points. Let $s = \sum_{i=1}^{m-1} \lambda_i = 1 - \lambda_m$. For $s > 0$, define

$$\bar{x} = \sum_{i=1}^{m-1} \frac{\lambda_i}{s} x_i.$$

Jensen's inequality: proof step 3

Step 1. The result holds for $m = 2$.

Step 2. Write $\sum_i \lambda_i x_i = s\bar{x} + \lambda_m x_m$.

Step 3. Two-point convexity gives

$$f(s\bar{x} + \lambda_m x_m) \leq sf(\bar{x}) + \lambda_m f(x_m).$$

Jensen's inequality: proof step 4

Step 1. The result holds for $m = 2$.

Step 2. Group the first $m - 1$ points into $\bar{x}$.

Step 3. Apply two-point convexity.

Step 4. The induction hypothesis gives

$$sf(\bar{x}) \leq s \sum_{i=1}^{m-1} \frac{\lambda_i}{s} f(x_i) = \sum_{i=1}^{m-1} \lambda_i f(x_i).$$

Adding $\lambda_m f(x_m)$ completes the proof.

□

Smoothness and Lipschitz gradients

Definition 12 (L -smoothness)

A differentiable function is L -smooth on C if

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \quad \forall x, y \in C.$$

If $f \in C^2$, then $\|\nabla^2 f(x)\|_2 \leq L$ implies L -smoothness. For a quadratic $f(x) = \frac{1}{2}x^\top Qx + b^\top x$, one may take

$$L = \lambda_{\max}(Q), \quad \mu = \lambda_{\min}(Q)$$

when $Q \succ 0$.

The descent lemma

Theorem 13 (Descent lemma)

If f has an L -Lipschitz gradient, then

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2.$$

This inequality supplies the standard upper quadratic model used to analyze gradient descent.

Descent lemma: proof step 1

Let $d = y - x$ and define $\phi(t) = f(x + td)$.

Step 1. The fundamental theorem of calculus gives

$$f(y) - f(x) = \int_0^1 \nabla f(x + td)^\top d \, dt.$$

Descent lemma: proof step 2

Let $d = y - x$.

Step 1. Write the objective difference as an integral.

Step 2. Add and subtract $\nabla f(x)$:

$$f(y) - f(x) = \nabla f(x)^\top d + \int_0^1 [\nabla f(x + td) - \nabla f(x)]^\top d \, dt.$$

Descent lemma: proof step 3

Step 1. Use the integral representation.

Step 2. Isolate the linear term.

Step 3. Cauchy–Schwarz and smoothness give

$$[\nabla f(x + td) - \nabla f(x)]^\top d \leq Lt \|d\|^2.$$

Descent lemma: proof step 4

Step 1. Use the integral representation.

Step 2. Isolate the linear term.

Step 3. Bound the integrand by $Lt \|d\|^2$.

Step 4. Integrating gives

$$f(y) - f(x) \leq \nabla f(x)^\top d + L \|d\|^2 \int_0^1 t \, dt = \nabla f(x)^\top d + \frac{L}{2} \|d\|^2.$$

This proves the claim. □

Strong convexity gives a lower quadratic model

For differentiable f , μ -strong convexity is equivalent to

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} \|y - x\|^2.$$

Consequences:

- at most one minimizer;
- quadratic growth: $f(x) - f^* \geq \frac{\mu}{2} \|x - x^*\|^2$;
- a well-defined condition number $\kappa = L/\mu$ for smooth strongly convex problems.

The Polyak–Lojasiewicz condition

Definition 14 (PL condition)

A differentiable function satisfies the PL condition with $\mu > 0$ if

$$\frac{1}{2} \|\nabla f(x)\|^2 \geq \mu(f(x) - f^*) \quad \forall x.$$

PL directly connects a computable local quantity, the gradient norm, to global suboptimality. PL does not require convexity and does not require a unique minimizer.

Strong convexity implies PL: proof step 1

Let x^* minimize a differentiable μ -strongly convex function.

Step 1. Strong convexity gives, for every y ,

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} \|y - x\|^2.$$

Strong convexity implies PL: proof step 2

Step 1. Use the strong convexity lower model.

Step 2. Minimize its right-hand side over y . The minimizer of the quadratic model is

$$y = x - \frac{1}{\mu} \nabla f(x).$$

Strong convexity implies PL: proof step 3

Step 1. Use the strong convexity lower model.

Step 2. Minimize the model at $y = x - \mu^{-1} \nabla f(x)$.

Step 3. Since $f^* \geq \min_y$ of the lower model,

$$f^* \geq f(x) - \frac{1}{2\mu} \|\nabla f(x)\|^2.$$

Strong convexity implies PL: proof step 4

Step 1. Use the strong convexity lower model.

Step 2. Minimize the model.

Step 3. Obtain $f^* \geq f(x) - (2\mu)^{-1} \|\nabla f(x)\|^2$.

Step 4. Rearranging yields

$$\frac{1}{2} \|\nabla f(x)\|^2 \geq \mu(f(x) - f^*),$$

which is the PL inequality. □

PL is strictly weaker than strong convexity

Consider $f(x, y) = x^2$ on $\mathbb{R}^2$. Then

$$\frac{1}{2} \|\nabla f(x, y)\|^2 = \frac{1}{2} \|(2x, 0)\|^2 = 2x^2 = 2(f - f^*).$$

Thus PL holds with $\mu = 2$.

However, $\nabla^2 f = \text{diag}(2, 0)$ has a zero eigenvalue. The function is not strongly convex, and every point $(0, y)$ is a global minimizer.

Key point. Strong convexity controls geometry in parameter space. PL only controls objective suboptimality through the gradient.

Constrained optimization problems

We write a general differentiable problem as

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & h_j(x) = 0, \quad j = 1, \dots, p, \\ & g_i(x) \leq 0, \quad i = 1, \dots, m. \end{aligned}$$

Equality constraints define surfaces. Inequality constraints define regions with boundaries. A point is feasible when it satisfies every constraint.

Active constraints and feasible directions

Definition 15 (Active constraint)

An inequality $g_i(x) \leq 0$ is active at a feasible x when $g_i(x) = 0$.

For differentiable constraints, a first-order feasible direction d satisfies

$$\nabla h_j(x)^\top d = 0, \quad \nabla g_i(x)^\top d \leq 0 \quad \text{for active } i.$$

Inactive inequalities impose no local first-order restriction. At a constrained local minimum, no feasible direction can satisfy $\nabla f(x)^\top d < 0$.

The Lagrangian collects objective and constraints

Definition 16 (Lagrangian)

For multipliers $\lambda \in \mathbb{R}^m$ and $\nu \in \mathbb{R}^p$,

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x).$$

The inequality multipliers satisfy $\lambda_i \geq 0$ for a minimization problem written with $g_i(x) \leq 0$. Equality multipliers are unrestricted.

KKT conditions

Theorem 17 (KKT necessity)

Under a suitable constraint qualification, every local minimizer x^ admits multipliers (λ^*, ν^*) satisfying:*

$$\text{stationarity: } \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0,$$

$$\text{primal feasibility: } g_i(x^*) \leq 0, \quad h_j(x^*) = 0,$$

$$\text{dual feasibility: } \lambda_i^* \geq 0,$$

$$\text{complementary slackness: } \lambda_i^* g_i(x^*) = 0.$$

For convex f, g_i and affine h_j , these conditions are sufficient for global optimality.

Equality multipliers: derivation step 1

Consider $\min f(x)$ subject to $h(x) = 0$, where the Jacobian $J_h(x^*)$ has full row rank.

Step 1. Feasible first-order directions lie in the nullspace:

$$J_h(x^*)d = 0.$$

Equality multipliers: derivation step 2

Step 1. Feasible directions satisfy $J_h(x^*)d = 0$.

Step 2. First-order optimality requires

$$\nabla f(x^*)^\top d = 0 \quad \text{for every } d \in \ker J_h(x^*).$$

Therefore $\nabla f(x^*)$ is orthogonal to $\ker J_h(x^*)$.

Equality multipliers: derivation step 3

Step 1. Feasible directions lie in $\ker J_h(x^*)$.

Step 2. $\nabla f(x^*)$ is orthogonal to this nullspace.

Step 3. Linear algebra gives

$$(\ker J_h)^\perp = \text{range}(J_h^\top).$$

Hence some ν^* satisfies $\nabla f(x^*) + J_h(x^*)^\top \nu^* = 0$.

Equality multipliers: derivation step 4

Step 1. Characterize feasible tangent directions.

Step 2. Apply first-order optimality on the tangent space.

Step 3. Use $(\ker J_h)^\perp = \text{range}(J_h^\top)$.

Step 4. Recognize the stationarity equation

$$\nabla_x \mathcal{L}(x^*, \nu^*) = \nabla f(x^*) + J_h(x^*)^\top \nu^* = 0.$$

This derives the Lagrange multiplier equation for equality constraints.



Complementary slackness selects active constraints

The product condition $\lambda_i^* g_i(x^*) = 0$ has two cases:

- if $g_i(x^*) < 0$, then $\lambda_i^* = 0$;
- if $\lambda_i^* > 0$, then $g_i(x^*) = 0$.

An inactive constraint has no local marginal effect. A positive multiplier identifies a binding constraint. An active constraint may still have a zero multiplier.

KKT sufficiency in convex problems

Theorem 18

Suppose f and g_i are convex, h_j are affine, and (x^, λ^*, ν^*) satisfies KKT. Then x^* is a global minimizer.*

KKT sufficiency: proof step 1

For any feasible x , convexity of the Lagrangian in x and stationarity give

$$\mathcal{L}(x, \lambda^*, \nu^*) \geq \mathcal{L}(x^*, \lambda^*, \nu^*) + \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*)^\top (x - x^*) = \mathcal{L}(x^*, \lambda^*, \nu^*).$$

KKT sufficiency: proof step 2

Step 1. Stationarity and convexity make x^* minimize the Lagrangian.

Step 2. For feasible x , $g_i(x) \leq 0$, $h_j(x) = 0$, and $\lambda_i^* \geq 0$, so

$$\mathcal{L}(x, \lambda^*, \nu^*) \leq f(x).$$

KKT sufficiency: proof step 3

Step 1. $\mathcal{L}(x, \lambda^*, \nu^*) \geq \mathcal{L}(x^*, \lambda^*, \nu^*)$.

Step 2. $\mathcal{L}(x, \lambda^*, \nu^*) \leq f(x)$ for feasible x .

Step 3. Primal feasibility and complementary slackness give

$$\mathcal{L}(x^*, \lambda^*, \nu^*) = f(x^*).$$

KKT sufficiency: proof step 4

Step 1. KKT makes x^* minimize the Lagrangian in x .

Step 2. Feasibility gives $\mathcal{L}(x, \lambda^*, \nu^*) \leq f(x)$.

Step 3. Complementary slackness gives $\mathcal{L}(x^*, \lambda^*, \nu^*) = f(x^*)$.

Step 4. Combining the inequalities,

$$f(x) \geq \mathcal{L}(x, \lambda^*, \nu^*) \geq \mathcal{L}(x^*, \lambda^*, \nu^*) = f(x^*).$$

Thus x^* is globally optimal.



The dual problem

Definition 19 (Dual function)

$$q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu).$$

The Lagrange dual problem is

$$\max_{\lambda, \nu} q(\lambda, \nu) \quad \text{subject to} \quad \lambda \geq 0.$$

The dual function is concave, even when the primal problem is not convex, because it is the pointwise infimum of affine functions of (λ, ν) .

Weak duality

Theorem 20 (Weak duality)

For every primal-feasible x and dual-feasible (λ, ν) ,

$$q(\lambda, \nu) \leq f(x).$$

Consequently, the dual optimum d^ never exceeds the primal optimum p^* .*

Weak duality: proof step 1

By definition of the infimum,

$$q(\lambda, \nu) = \inf_z \mathcal{L}(z, \lambda, \nu) \leq \mathcal{L}(x, \lambda, \nu).$$

Weak duality: proof step 2

Step 1. $q(\lambda, \nu) \leq \mathcal{L}(x, \lambda, \nu)$.

Step 2. For primal-feasible x and $\lambda \geq 0$,

$$\sum_i \lambda_i g_i(x) \leq 0, \quad \sum_j \nu_j h_j(x) = 0.$$

Therefore $\mathcal{L}(x, \lambda, \nu) \leq f(x)$.

Weak duality: proof step 3

Step 1. $q(\lambda, \nu) \leq \mathcal{L}(x, \lambda, \nu)$.

Step 2. Feasibility gives $\mathcal{L}(x, \lambda, \nu) \leq f(x)$.

Step 3. Hence $q(\lambda, \nu) \leq f(x)$. Maximizing the left side and minimizing the right side yields

$$d^* \leq p^*.$$

This proves weak duality.



Strong duality and Slater's condition

Theorem 21 (Strong duality for convex programs)

For a convex problem with affine equalities, if a strictly feasible point exists,

$$g_i(\bar{x}) < 0, \quad h_j(\bar{x}) = 0,$$

then the primal and dual optimal values coincide: $p^ = d^*$, and dual optimal multipliers exist under mild regularity assumptions.*

Strict feasibility is known as Slater's condition. It also provides the constraint qualification needed for KKT necessity in this setting.

Dual multipliers measure sensitivity

Let $p(u)$ denote the optimal value after relaxing a constraint to $g_i(x) \leq u_i$. Under regularity and differentiability of the value function,

$$\frac{\partial p}{\partial u_i}(0) = -\lambda_i^*.$$

Thus a large positive λ_i^* means that a small relaxation decreases the optimal cost rapidly. For a resource constraint, the multiplier acts as a shadow price.

Example: constrained regression

Consider least squares with a budget on coefficients:

$$\min_w \frac{1}{2} \|Xw - y\|^2 \quad \text{subject to} \quad \|w\|_2^2 \leq R^2.$$

The Lagrangian is

$$\mathcal{L}(w, \lambda) = \frac{1}{2} \|Xw - y\|^2 + \lambda(\|w\|^2 - R^2), \quad \lambda \geq 0.$$

Stationarity gives

$$(X^\top X + 2\lambda I)w = X^\top y.$$

When the budget binds, the multiplier produces a ridge-like regularization strength.

Example: simplex constraints

Portfolio weights and mixture probabilities often satisfy

$$x \in \Delta_d, \text{quad} x_i \geq 0, \text{quad} \mathbf{1}^\top x = 1.$$

For a convex objective $f(x)$, write inequalities as $-x_i \leq 0$:

$$\mathcal{L}(x, \lambda, \nu) = f(x) - \lambda^\top x + \nu(\mathbf{1}^\top x - 1).$$

KKT stationarity is $\nabla f(x) - \lambda + \nu \mathbf{1} = 0$, with $\lambda_i x_i = 0$. Positive components share the same adjusted marginal objective.

Example: soft-margin SVM

The primal problem is

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \\ \text{subject to} \quad & 1 - \xi_i - y_i(w^\top x_i + b) \leq 0, \\ & -\xi_i \leq 0. \end{aligned}$$

KKT conditions identify support vectors through positive multipliers. The dual depends only on inner products $x_i^\top x_j$, which permits the kernel construction.

Summary

- Convex combinations define convex sets and functions.
- Gradients and Hessians provide equivalent tests for smooth convex functions.
- Smoothness gives an upper quadratic model; strong convexity gives a lower one.
- Strong convexity implies PL, but PL allows multiple minimizers and nonconvex functions.
- KKT combines stationarity, feasibility, multiplier signs, and complementary slackness.
- Dual variables provide bounds and local sensitivity information.